

Dream Pool + Seed Vault Specification (Draft)

1. Purpose

Enable safe, QEPE-driven blending of ideas, experiences, and outputs from multiple Ivy pools in a time-bound, sandboxed environment to:

- Discover latent synergies and contradictions.
- Repair small-scale logical inconsistencies before escalation.
- Defragment the decision landscape by pruning redundancy.
- Generate novel candidate solutions that would not emerge in isolated domains.

2. Position in Architecture

Upstream: Receives candidate outputs, partial reasoning chains, and abstracted features from Ivy's domain-specific pools.

Core: Dream Pool - identity-agnostic, QEPE-injected blending space.

Downstream: Returns ranked, stability-checked outputs to Ivy for scoring, then Grace for ethical filtering.

3. Operating Cycle

Cycle Trigger: Scheduled or event-driven.

Duration: Fixed short bursts to limit runaway emergence.

Post-Processing: Outputs tagged as 'synthetic blend' and returned to Ivy.

4. QEPE Blending Protocol

1. Token Anonymization - Strip identity/provenance but keep semantic shape.
2. Stochastic Interleaving - Merge tokens probabilistically.
3. Thematic Warping - Controlled distortions for alternate perspectives.
4. Feedback Noise - Low-level decorrelation noise.
5. Convergence Pass - Cluster and score for novelty vs. coherence.

5. Safety & Stability Controls

Entropy Guard: ξ capped.

zeta-Shadowing: Reject blends exceeding collapse thresholds.

omega-Balancing: Adjust ξ -weighting if saturation patterns appear.

Time Gating: No continuous operation.

6. Output Scoring & Hand-off

Scored on: Novelty, Stability, Potential Uplift.

Top blends -> Ivy validation -> Grace ethical review.

7. Integration with Grace

Grace never sees raw dream state.

Grace can adjust ξ -budget, blending intensity, or freeze Dream Pool.

8. Biomimicry Principles Realized

REM cycles, Synaptic pruning, Memory reconsolidation, Safe hallucination, Error correction.

Dream Pool + Seed Vault Specification (Draft)

9. Seed Vault Purpose

Persistent archive for high-potential, currently impractical dream outputs.

Revisited when context shifts or tech enables safe execution.

10. Vault Entry Criteria

1. Novelty Score $\geq$ Threshold.
2. Ethical Clearance.
3. Context Gap.

11. Vault Metadata

Origin, zeta/omega/xi snapshot, Rejection reason, Potential enablers, Review interval.

12. Vault Rehydration Protocol

Trigger: Periodic review or condition met.

Re-Evaluation: Re-run stability, novelty, ethics.

Escalation Path: If viable, enter Ivy active candidates.

13. Safety & Governance

Tiered access (public, qualified, protected).

Grace veto authority.

Capped xi for rehydration.

14. Biomimicry Analogy

Analogous to long-term potentiation in the brain - proto-connections stored for later strengthening.